# MultiDiffusion

While diffusion models have revolutionized image synthesis quality, achieving precise user control and task adaptation typically requires a new training of the model with the new adaption, resource-intensive fine-tuning or specialized architectural modifications. We want to have a flexible way to generate images. Some examples of the application could be:

- Generate an image with **arbitrary aspect ratio**. Normally the models have a fixed square output's resolution.
- Control where objects **are located** (spatial control)

We want to achieve these goals and others without training the model from scratch or perform time consuming methodology in order to adapt the model on the specific task.

To bypass these limitations, we introduce **MultiDiffusion**: a unified framework that leverages pre-trained models to provide versatile image manipulation. The approach used in the paper there is a new generation process, based on an optimization problem. MultiDiffusion synchronizes multiple generation paths under a single set of constraints. This allows for high-fidelity synthesis across various user-defined controls, including custom aspect ratios like panoramas and spatial layouts defined by segmentation masks or bounding boxes.

## Introduction

When we want to deploy text-to-image models to real-world applications, we have to face some difficulty due to the lack of intuitive control over the content that is generated.
Some common solutions are:

1) Training a model from scratch to specialize the generator on the new task. Control the generation using additional guiding signal during the training (i.e. semantic layout, depth map, …). This approach is very expensive and time consuming.
2) Finetune the diffusion model for the task at hand (i.e. inpainting, layout-to-image training).
3) Involve a pre-trained model and add on some controlled capability.

Every methods have different cons however

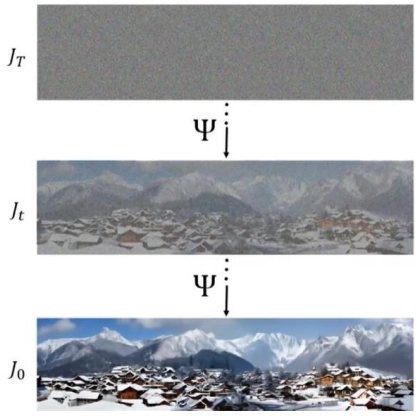
The MultiDiffusion approach doesn't need a training and it's a general flexible way to control the output of the model.

The MultiDiffusion defines a new generation process which is composed of several generation binded together with a set of constraints (or shared parameters)

# MultiDiffusion process

In order to continue the explanation we consider a panorama as the final result that we want achieve.

*Example* (sample a panorama instead of an image):
Prompt conditioning: “a photo of snow-capped mountain village”




$\Psi$: multi diffusion process
$J_i$ : panorama at time step i

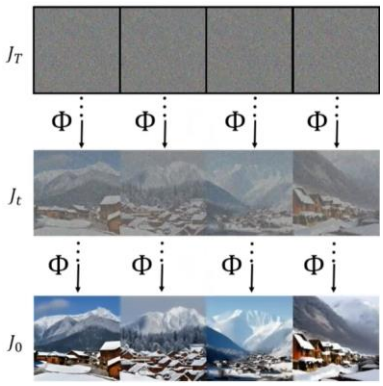
Steps:

1) We start with a noise gaussian panorama image sampling
2) Iterative denoise the panorama using the MultiDiffusion process
3) Reach the seamless and high quality panorama (matching the input text prompt)

We don’t have a text-to-panorama model and we’d like to use the pre-trained diffusion model that we have in order to not train everything again (it works on fixed-resolution images)

### Naïve approach

A simple idea is to separate the panorama on individual crops.




$\Phi$: *pre-trained diffusion model that works with a fixed-resolution images*

Steps:

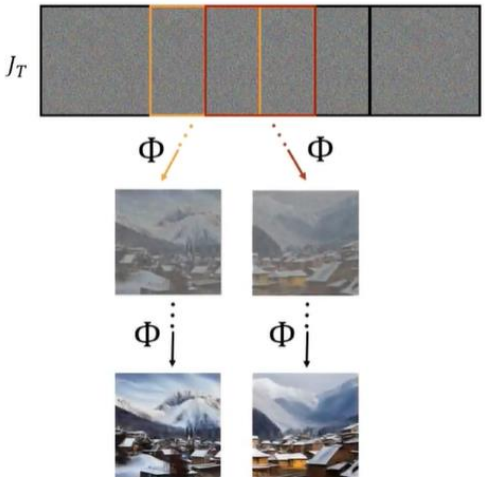
1) Divide the panorama on different individual crops
2) Apply the pre-trained diffusion model in each crop
3) Denoise the crops until we get the final images
4) Merge the crops to get the panorama (merged image)

We can see that we don’t get a seamless and consistent panorama because each one of the crop will lead to different diffusion process

### Overlapped crops

Another solution consist to overlap the crops in order to unify the regions between the images.

*Example* (overlapped crops):

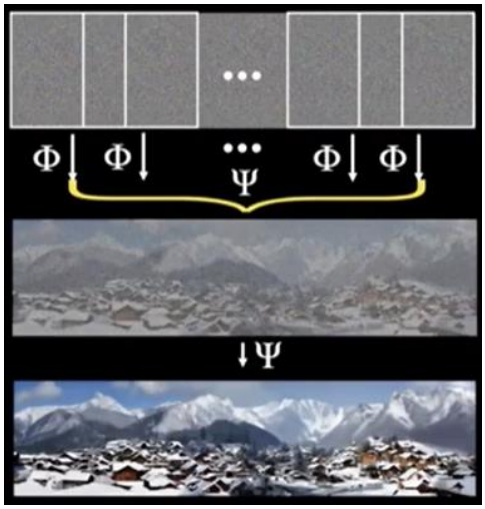



Even with overlapping crops, the diffusion process will end (after the denoising step) with non-consistent images. The overlapped pixels don't match because because each of these images takes a different trajectory in the diffusion process. So, these processes are just two separated generations: they might have started form similar point but they diverge to different generations.

### Fusing diffusion paths

The idea behind MultiDiffusion is to fuse the paths of the overlapped crop images. In other words, the goal is to fuse the independent generation processes into one unified global denoising step.

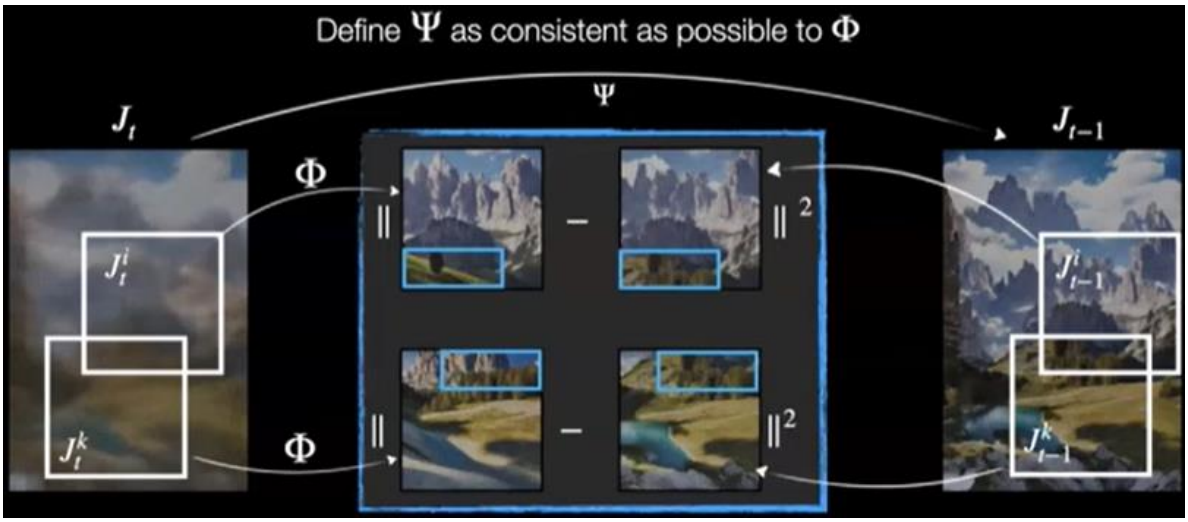
*Example* (MultiDIffusion process):




Steps:

1) Divide the panorama (input noise) into various **overlapping regions**
2) Apply (at each denoising step) the **diffusion model** ($\Phi$) independently on each crop (may pull to different direction).
3) Consolidate all these inconsistent prediction using MultiDIffusion model ($\Psi$) in order to define one **global denoising step** for the entire panorama
4) Apply $\Psi$ iteratively until we reach the final clean panorama

## MultiDiffusion function




$\Phi$: pre-trained diffusion model
$\Psi$: it performs the global denoising step (it takes as input $J_t$)
$J_t$: panorama image at time step t (input)
$J_{t-1}$: panorama image at time step t-1. It is less noisy version of $J_t$ (output)

As we said before, even though we're taking these crops ($J^k$ and $J^i$) from the same region (with overlapping parts) the denoising prediction, when we apply the pre-trained diffusion model, isn't longer going to respect the overlap. In the end we'll get different solutions (diffusion model is being applied independently on each crop).

The constraint that we want to have is that the overlapping regions have the **same pixels value** (same content).

In practice, we take the average on overlapping regions

The cost function is defined as follow:

$$J_{t-1} = \Psi(J_t) = \text{argmin}_J \sum_i \|\Phi(J_t^i) - J^i\|^2$$

### Core mechanism

Instead of generating an entire image in a single pass, MultiDiffusion applies a reference diffusion model (denoted as $\Phi$) to different regions or "crops" of a target image.

1. **Mapping and Denoising**: the process defines a set of mappings, that relate the target image space to the reference model's space (e.g., taking square crops from a larger panoramic canvas). For each region, the reference model predicts a denoising step based on a specific text prompt or condition.
2. **Global Reconciliation**: Because these overlapping regions may pull in different denoising directions, MultiDiffusion reconciles them into a single, global denoising step.
3. **Least-Squares Optimization**: This reconciliation is formulated as an optimization task (FTD). The FTD loss reconciles the different denoising sampling steps.

For many practical applications like panoramas, this optimization can be solved analytically. Each pixel in the new image is essentially a weighted average of all the diffusion updates that overlap at that pixel's location